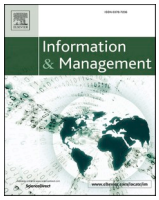
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Self-disclosure in online social networks: The needs-affordances-features perspective

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ABSTRACT

Drawing on affordance theory and the needs–affordances–features (NAF) perspective, this paper examines the impacts of online social network (OSN) affordances and psychological ownership on users' self-disclosure. Through the analysis of data collected from a field survey and a scenario-based experiment, we found that accessibility, information retrieval, editability, and association affordances positively affect psychological ownership, which further determines users' self-disclosure on OSNs. This paper not only advances the understanding of OSN self-disclosure by emphasizing the significance of OSN affordances and psychological ownership but also offers practical insights for OSN providers to enhance self-disclosure by designing effective affordances.

1. Introduction

The usage of online social networks (OSN), such as WeChat, TikTok, and Facebook, has seen a significant surge in the past decade [1]. The global OSN user base has now reached 3.1 billion, with individuals spending an average of 2.5 h daily on social media and microblogging services [2]. Despite the substantial growth of the OSN industry, a persistent challenge for OSN platforms is to enhance users' *self-disclosure*, which refers to voluntary and intentional personal information disclosure to others on the platform [3]. As users' online self-disclosure serves as a crucial data source that provides valuable insights for OSN platforms to retain their users, platforms have continually implemented or upgraded technologies such as cookies and clickstream technologies to gather data on user online behavior [4].

Given its significance, OSN self-disclosure has been extensively examined from two distinct perspectives in the information systems (IS) research community. The dominant *economic* perspective draws on theories of privacy calculus and social exchange to construe OSN self-disclosure as a rational process by which users with stable preferences for privacy weigh the expected privacy benefits against privacy risks [1,5–16]. More recently, the *behavioral* perspective has leveraged the principle of bounded rationality to frame OSN self-disclosure as a bounded rational process in which consumers' cognitive biases and emotional reactions occur predictably [3,17–23]. The core assumption of both research paradigms is that OSN self-disclosure is determined by generic sociopsychological factors, such as privacy benefits, privacy risks, and emotions [24,25].

While these perspectives offer valuable insights, the assumption that OSN self-disclosure stems from generic sociopsychological factors has faced questioning in IS research [18,23]. IS scholars recommend a shift to emphasizing contextual technological features over generic sociopsychological factors in OSN self-disclosure studies. First, critiques within IS research highlight the significance of and advocate considering these contextual technological features when investigating self-disclosure behaviors on OSNs [5,26,27]. Furthermore, Hong et al. [28, p.111] contend that "the economic/behavioral research paradigm is passive with respect to technology, often ignoring or under-theorizing the IT artifact itself." Second, in real-world business scenarios, technological advancements within OSNs play a crucial role in shaping user behavior, particularly users' self-disclosure [16]. However, existing research has examined the impact of specific technological features inadequately, such as friending and contacting capabilities, search and privacy settings, post and editing functionalities, and relational tie mechanisms, on user self-disclosure behavior in OSNs. This lack of investigation hinders OSN platforms from leveraging fast-growing technologies to improve user self-disclosure behavior.

To fill the identified research gap, a valuable theoretical lens for contextually theorizing technological features is *technology affordances*. These affordances refer to the goal-oriented action possibilities enabled by the technological features of technical objects for specific user groups

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[29]. OSNs encompass multiple technological features, providing different technology affordances, which are crucial predictors influencing user self-disclosure behavior on OSN platforms. Specifically, OSNs, as technical objects, align with four types of technology affordances [30] derived from Kane et al.'s [30] social network research framework: (1) *accessibility affordance*—the OSN platform offers users the action possibility to connect with other users; (2) *information retrieval affordance*—the OSN platform offers users the action possibility to gather information about other users; (3) *editability affordance*—the OSN platform offers users the action possibility to erase, edit, and hide their posted content; and (4) *association affordance*—the OSN platform offers users the action possibility to associate the responsibility for their posted information with other users with whom they share a connection.

The *needs–affordances–features (NAF)* perspective explains how individuals' innate psychological needs drive their use of social media applications, depending on how well these applications offer features and affordances to satisfy those needs (Karahanna et al. [31]). An important psychological need underlying the impacts of technology affordances on OSN self-disclosure is the formation of *psychological ownership*, which captures the state of ownership of technical objects arising from need for having a place, need for self-identity, and need for efficacy and effectance [31]. OSNs provide users with multiple technology affordances enabling social activities, such as accessing other users, retrieving other users' information, editing the posted information, and associating the posted comments with other users [32]. Through these activities, the OSN platform transforms into a virtual territory for users to express their self-identity and self-capability [33]. Users are likely to disclose their personal information when they establish a state of psychological ownership by viewing the OSN platform as their own virtual territory [10]. However, the precise mechanism by which technology affordances affect OSN self-disclosure through the establishment of psychological ownership remains poorly understood in the current literature.

To bridge the aforementioned backdrop, we propose a contextualization methodology for identifying and challenging assumptions underlying the existing OSN self-disclosure literature. We argue that, in addition to generic sociopsychological factors, understanding the impacts of contextual technological features is needed in the OSN self-disclosure literature. Hong et al. [28] suggest two contextual theorizing approaches: (1) *single-context theory contextualization*, which focuses on incorporating contextual factors as antecedents of endogenous variables, and (2) *cross-context theory contextualization*, which centers on replicating theoretical models in different contexts. Following the single-context theory contextualization approach, we develop a model to examine contextual technological features as antecedents of users' OSN self-disclosure. First, we draw on affordance theory [34] and identify four types of OSN affordances that are closely tied to contextual technological features of the OSN platform: accessibility, information retrieval, editability, and association affordances. Second, we draw on the NAF perspective [31] to hypothesize that these OSN affordances positively influence self-disclosure through the formation of users' psychological ownership.

To examine the research model rigorously, we employ a multimethod approach that comprises a field survey and a scenario-based experiment. In Study 1, a field survey is conducted to test the research model in a natural setting, aiming to maximize external validity. In Study 2, a scenario-based experiment is employed to cross-validate the research model, allowing the establishment of causal inferences and the enhancement of internal validity. A multimethod approach is essential because it provides a comprehensive understanding of the phenomenon, offering both correlational and causal perspectives that lead to more robust inferences than using a single method. Moreover, the multimethod approach strengthens the external and internal validity of our findings. With such additional information and evidence, we better explain, refute, or provide complementary knowledge to existing theories.

2. Theoretical background

2.1. OSN self-disclosure

Self-disclosure refers to the extent to which users voluntarily and intentionally disclose personal information to others. It has the following dimensions: (1) *amount*—the frequency and duration of users' information disclosure; (2) *depth*—the degree of intimacy of users' information disclosure; (3) *honesty*—the degree of accuracy and truth with which users disclose personal information; (4) *intent*—users' perceived control and awareness of their information disclosure; and (5) *valence*—the degree of positivity of users' information disclosure [35]. Accordingly, we define *OSN self-disclosure* as the degree of amount, depth, honesty, intent, and valence with which users voluntarily and intentionally reveal their personal information on social networking sites. Our literature review reveals two alternative perspectives on prior OSN self-disclosure studies: *economic* and *behavioral*.

The *economic* perspective has leveraged the principle of economic man to view OSN self-disclosure as a rational trade-off between expected privacy benefits and their potential privacy costs. For instance, privacy calculus theory is the most popular theory used to explain OSN self-disclosure [1,6–9]. Researchers have found that OSN self-disclosure is an interplay between their expected social benefits and costs, such as self-presentation, relationship bonding, privacy risks, and privacy concerns. Social exchange theory has also been employed to explain how OSN self-disclosure is determined by the overall worth of a social information exchange based on privacy benefits and costs, such as social capital bonding, social capital bridging, interpersonal awareness, and privacy concerns [5,12–14,16]. Likewise, some studies based on communication privacy management theory contend that privacy assurance approaches can enhance privacy control and alleviate privacy concerns, therefore facilitating OSN self-disclosure [10,15,36].

The *behavioral* perspective has built upon the principle of bounded rationality to construe OSN self-disclosure as a bounded rational process in which users' cognitive biases and emotional reactions predictably occur. For instance, some researchers used affect heuristic theory to explain how users' emotional reactions influence their decision to disclose personal information on the OSN platform [23,37]. Some studies draw on the five-factor model of personality to investigate the impacts of agreeableness, conscientiousness, extraversion, neuroticism, and openness on OSN self-disclosure [20–22]. Similarly, some researchers build upon coping theory to illustrate how OSN self-disclosure serves as a means for users to escape from real-life worries and problems, such as loneliness, stress, role conflict, and role overload [3,17,19].

Hence, our literature review makes two observations. First, most studies have primarily examined the antecedents of OSN self-disclosure pertaining to generic sociopsychological factors, leaving contextual technological features (i.e., OSN affordances) largely underexplored. Second, limited studies have investigated the mediating psychological mechanism (i.e., psychological ownership) by which OSN affordances affect OSN self-disclosure. To address these research gaps, our paper employs the *behavioral* perspective to investigate the effects of OSN affordances on OSN self-disclosure, along with examining the mediating role of psychological ownership.

2.2. Affordance theory

Affordance theory has been widely used in the IS discipline to theorize the effects of contextual technological features on users' technology use [31,32,38–40]. The concept of *technology affordances* refers to "the potential for behaviors associated with achieving an immediate concrete outcome and arising from the relation between a technical object and a goal-oriented user" [41, p. 832]. Prior IS studies conceptualize technology affordances from two key perspectives: *relational* and *multiple*. The *relational* perspective highlights that technology affordances are the goal-oriented action possibilities afforded by the

technological features of a technical object for a specified user [29]. In this view, the objective technological features of a technical object are necessary conditions for technology affordances, but not the technology affordances themselves. Moreover, technology affordances are action possibilities arising from the relations between a technical object and a specific user, which have to be perceived subjectively by the user. The *multiple* perspective emphasizes that a technical object often has multiple technological features, and those technological features afford different action possibilities, thereby offering multiple technology affordances to users [42]. Norman [34] argues that good designers purposefully build multiple affordances into a technical object to suggest how its technological features should be used.

Accordingly, we define *OSN affordances* as the goal-oriented action possibilities afforded by the technological features of OSNs for a specific user. OSNs have multiple technological features and therefore offer users multiple technology affordances [31]. Kane et al. [30] propose a taxonomy of OSN technological features with four archetypes: (1) *network transparency*—the OSN platform allows users to *access and traverse their connections* (e.g., friending, contacting) and those made by others; (2) *search and privacy*—the OSN platform enables users obtain digital content of others and protect it from various *information retrieval mechanisms* (e.g., searching); (3) *digital profile*—the OSN platform provides a *unique editable user profile* (e.g., user profile) that is constructed by the user, by members of their network, and by the platform; and (4) relational ties—the OSN platform offers various *association mechanisms* (e.g., @username, #hashtags) for users to articulate a list of other users with whom they share a connection. Based on Kane et al.'s [30] topology of OSN technological features and following Chan et al.'s [38] approach to developing technology affordances, we identified four types of OSN affordances for self-disclosure: (I) accessibility affordance, (II) information retrieval affordance, (III) editability affordance, and (IV) association affordance (see Table 1).

Accessibility affordance refers to the extent to which OSNs offer users action possibilities to connect to other users on a platform. This affordance was labeled by prior studies as accessibility affordance [43], allocentric affordance [31], and network transparency [30]. Accessibility affordance enables users to transcend time and spatial constraint in connecting to others for the purpose of effective information disclosure through OSN network transparency features, such as friending, following, and contacting. For instance, Fox and Moreland [43] highlight accessibility affordance as the action possibility offered by a social networking site for users to connect to others conveniently. Likewise, Karahanna et al. [31] contend that users can access other users easily through forming relationships and communicating with other users on a social networking site.

Information retrieval affordance refers to the extent to which OSNs offer users action possibilities to obtain information about other users on a platform. This affordance was tagged by prior studies as persistence affordance [39,43–45], visibility affordance [39,43–45], allocentric affordance [31], and search and privacy [30]. Information retrieval affordance allows users to obtain personal information shared by others in order to understand their backgrounds, preferences, and daily activities for the purpose of reciprocal information disclosure through OSN search and privacy features, such as searching and browsing. For instance, Chen et al. [39] and Evans et al. [44] emphasize accessibility affordance as the action possibility offered by an enterprise social medium that makes employees' behavior, preferences, and network connections available to others. Similarly, Fox and Moreland [43] and Raja-Yusof [45] suggest that users can retrieve information on others through browsing others' profiles, albums, and network connections on social networking sites.

Editability affordance refers to the extent to which OSNs offer users action possibilities to manipulate the content that they have posted, commented on, and shared on the platform. This affordance was identified by prior studies as editability affordance [39,44,45], egocentric

Table 1
OSN affordances.

Affordance	Definition	How the affordance relates to self-disclosure	Related OSNs affordances
Accessibility	The extent to which OSNs offer users action possibilities to connect to other users on the platform.	This affordance enables users to transcend time and spatial constraint in connecting to others for the purpose of effective information disclosure through OSN network transparency features, such as friending, following, and contacting.	Accessibility affordance [43]; allocentric affordance [31]; network transparency [30]
Information retrieval	The extent to which OSNs offer users action possibilities to obtain information about other users on the platform.	This affordance allows users to obtain personal information shared by others to understand their background, preferences, and daily activities for the purpose of reciprocal information disclosure through OSN search and privacy features, such as searching and browsing.	Persistence affordance [39,43–45]; visibility affordance [39, 43–45]; allocentric affordance [31]; search and privacy [30]
Editability	The extent to which OSNs offer users action possibilities to manipulate comments that they posted, commented on, and shared on the platform.	This affordance empowers users to manipulate their own information and posted comments for the purpose of secure information disclosure through OSN digital profile features, such as erasing, editing, and hiding.	Editability affordance [39,44,45]; egocentric affordance [31]; digital profile [30]
Association	The extent to which OSNs offer users action possibilities to associate the responsibility for their posts with other users who interacted with the posts on the platform.	This affordance authorizes users to articulate lists of other users with whom they share a connection for the purpose of interactive information disclosure through OSN relational tie features, such as directed associations ("@") and hashtags ("#").	Association affordances [39,44]; network-informed associating [46,47]; collective affordance [32,45,48]; allocentric affordance [31]; relational ties [30]

affordance [31], and digital profile [30]. Editability affordance empowers users to manipulate their profile information and posted comments for the purpose of secure information disclosure through OSN digital profile features, such as erasing, editing, and hiding. For instance, Chen et al. [39] and Evans et al. [44] conceptualize editability affordance as the action possibility for employees to revise, add to, and modify their posts in enterprise social media. Likewise, Raja-Yusof [45] argues that editability affordance allows users to update their user profiles and posted comments on a social networking site.

Association affordance refers to the extent to which OSNs offer users action possibilities to associate the responsibility for their posted information with others who interacted with the posted information on the platform. This affordance was examined by prior studies as association affordance [39,44], network-informed associating [46,47], collective affordance [32,45,48], allocentric affordance [31], and relational ties [30]. Association affordance authorizes users to articulate a list of other users with whom they share a connection for the purpose of interactive information disclosure through OSN relational tie features, such as directed associations ("@") and hashtags ("#"). For instance, Chen et al. [39] and Evans et al. [44] operationalize association affordance as the action possibility for employees to associate their posts with other organizational members who interact with the posts in the enterprise social media. Similarly, Majchrzak et al. [46] and Sæbø [47] highlight that network-informed associating affordance allows employees to engage in the knowledge conversion informed by relational and content ties in the enterprise social media.

2.3. The needs–affordances–features perspective

The NAF perspective, an IS variance of psychological ownership theory, has emerged to elucidate how individuals' innate psychological needs motivate their use of social media applications to the extent to which the applications provide features and affordances to fulfill these needs (Karahanna et al. [31]). Building on this framework, we can elucidate the motivations behind users' engagement with OSN applications (see Fig. 1). Specifically, the OSN platform has multiple features and affordances (e.g., OSN affordances) that can fulfill certain psychological needs of individuals (e.g., psychological ownership), which further motivate their various uses of the platform (e.g., self-disclosure behavior). For example, WeChat, a popular OSN platform in China, provides functions such as "messages" or "voice or video call," enabling users to connect with others. Leveraging these affordances can satisfy individuals' psychological need for relatedness, which in turn drives them to use the WeChat applications for disclosing personal information.

We have selected *psychological ownership* as the needs-related construct to assist in identifying the salient psychological needs fulfilled by using online social networks. This choice stems from the notion that individuals tend to develop *a feeling of ownership* when they devote effort to generating content on an online social network platform [49].

Psychological ownership refers to "the state in which individuals feel as though the target of ownership or a piece of that target is theirs" [49, p. 86]. Individuals establish psychological ownership toward an object when they have a possessive connection the object and view it an extension of themselves [50–52].

According to psychological ownership research, individuals form psychological ownership from three basic psychological needs: need for having a place, need for self-identity, and need for efficacy and effectance [49]. *Need for having a place* refers to an individual's innate psychological need to have a certain territory or space that can serve as a "home" place [49]. Individuals devote significant energy and resources in seeking spaces that can potentially become their homes and offer them feelings of comfort. *Need for self-identity* refers to an individual's innate psychological need to have a clear sense of self and distinguish her/himself from others [49]. Need for self-identity comprises three distinct aspects: (1) coming to know the self—an individual's psychological need to learn about her/himself through self-definition and self-knowledge; (2) expression of self-identity—an individual's psychological need to express her/his personal self-identity to others; and (3) maintaining the continuity of self-identity—an individual's psychological need to connect personal self-identity emotionally with her/his past. *Need for efficacy and effectance* refers to an individual's psychological need to have the capability of controlling and altering the environment around her/himself [49]. The need to experience efficacy and effectance can lead to psychological ownership of a variety of objects that can even come to be considered part of the extended self.

3. Research model and hypotheses

Fig. 2 presents the research model and research hypotheses. Drawing upon affordance theory [34] and the NAF perspective [49], we postulate that OSN affordances influence users' psychological ownership positively, which further affects their self-disclosure behavior. In particular, age, gender, education, income, usage experience, usage frequency, and privacy concerns are included as control variables in the research model.

3.1. Psychological ownership and self-disclosure

Psychological ownership captures users' feelings of ownership toward OSNs, which comprise need for having a place, need for self-identity, and need for efficacy and effectance [49]. OSNs offer users a virtual "home" space in which they can join conversations with their family members or intimate friends [31]. Users can voluntarily manipulate their OSN homepages with colorful backgrounds and photos [13]. Also, users can use photos, videos, and dairies posted on their OSN homepages to express self-identity and maintain self-continuity, in that they are concrete reminders of who they were [33]. Therefore, users easily develop feelings of psychological ownership toward OSNs and view OSN homepages as their possessions [53]. Once psychological ownership toward OSNs is established, users are more likely to reveal

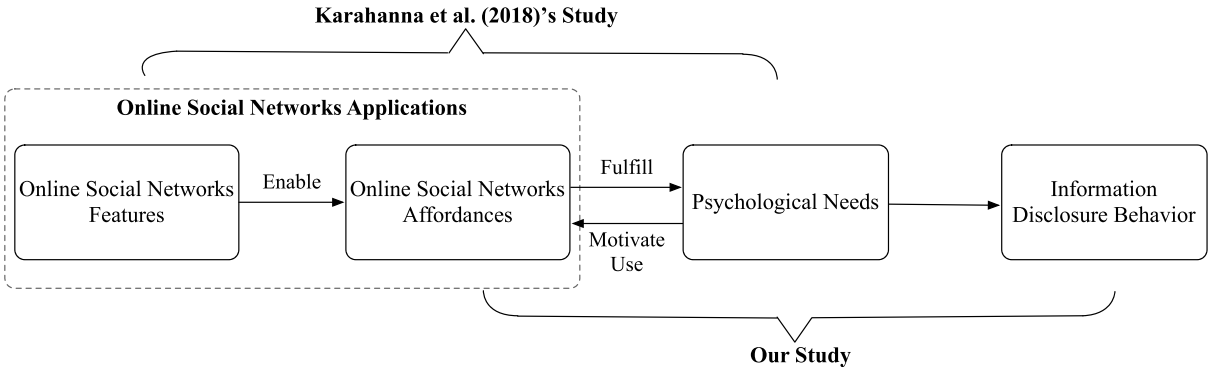


Fig. 1. Logic of the NAF perspective (adapted from Karahanna et al. [31]).

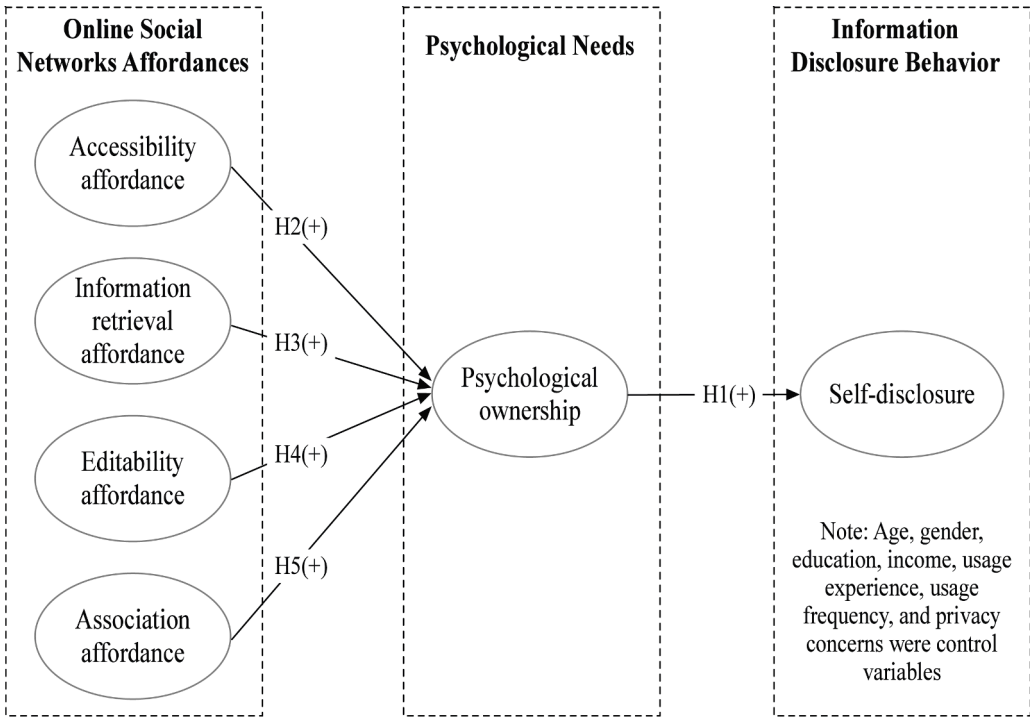


Fig. 2. Research model.

personal information because OSNs offer them a "home" territory to express their self-identities and self-capabilities through information disclosure. Therefore, we hypothesize the following:

H1. *Psychological ownership influences user self-disclosure positively.*

3.2. Accessibility affordance and psychological ownership

Accessibility affordance in OSNs refers to the technological features that enable users to transcend temporal and spatial constraints, allowing them to send friend requests or engage in chats with other users regardless of time or location [54]. First, the accessibility affordance directly addresses the need for having a place. By enabling constant connectivity and interaction, this affordance allows users to cultivate a sense of familiarity and stability within OSNs. As users repeatedly engage with the platform and establish social connections, the OSN becomes a virtual "home" that satisfies their need for having a place, thereby strengthening their psychological ownership of the platform [55,56]. Second, the accessibility affordance contributes significantly to fulfilling the need for self-identity. This affordance equips users with tools for making connections and interacting with like-minded individuals [31], enabling them to engage easily and continuously in activities that express their opinions and self-identity within the OSN [57,58]. The continual access to these self-affirming activities allows users to maintain a consistent and coherent self-identity over time [10,51,59]. Third, the accessibility affordance addresses users' need for efficacy and effectance directly. This affordance empowers users to navigate OSN functionalities easily, such as building relationships and communication, boosting their perceived competence and control within the OSN environment [60]. When users experience smooth platform interaction and achieve their desired outcomes seamlessly, their sense of efficacy is reinforced. This perceived competence and control over the OSN environment play a pivotal role in fostering psychological ownership. Therefore, we hypothesize the following:

H2. *Accessibility affordance influences psychological ownership positively.*

3.3. Information retrieval affordance and psychological ownership

Information retrieval affordance in OSNs captures the technological features that allows users to search for, locate, and retrieve other users' information on the platform, including profiles, preferences, behavior, and social connections [38]. First, this affordance directly addresses the need for having a place. The ability to retrieve relevant information easily on an OSN enables users to organize and personalize their interactions and content-sharing activities. As users consistently discover and curate information that resonates with their interests and values, the OSN transforms into a personalized and familiar environment—analogous to a digital "home." This personalized experience and the effortless access to preferred content contribute to fulfilling users' psychological need for having a place within the OSN [55]. Second, this affordance fulfills the need for self-identity by enabling users to obtain and compare information about other users. Within OSNs, users not only share their personal feelings and information, but also expect to receive similar information in return [23]. This reciprocal information exchange allows users to engage in social comparison, providing them with insights into others' perceptions of them and their self-perception [31,61]. Through this process, users can better understand and express their self-identity, and the continuous engagement with identity-relevant information also helps users maintain the continuity of their self-identity over time, further enhancing their psychological ownership of the OSN. Third, this affordance significantly influences the need for efficacy and effectance by enabling users to explore and learn their skills and competencies and compare them with those of others on the OSN platform [11,35]. As users navigate the platform and engage in these comparative processes, they gain a deeper understanding of their capabilities and the impact of their actions. This sense of competence and control reinforces their psychological ownership, as users feel empowered by their ability to influence their environment and achieve desired outcomes within the OSN [60]. Therefore, we hypothesize the following:

H3. *Information retrieval affordance influences psychological ownership positively.*

3.4. Editability affordance and psychological ownership

Editability affordance in OSNs captures the technological features that offer users the possibility of modifying, updating, and refining their content before or after it is posted [38], thereby allowing them to manipulate personal feelings and information. First, this affordance significantly addresses the need for having a place by empowering users to customize their digital environment according to their preferences, including purposeful information sharing and targeted disclosure to specific audiences [39,44]. The ability to adjust content over time, whether refining a post, updating a profile, or editing a comment, enables users to align their online presence with their evolving needs and preferences. Such personalization strengthens users' attachment to the platform, making it feel more like "their own place" and thereby deepening their psychological ownership [55]. Second, it contributes significantly to fulfilling the need for self-identity by empowering users to craft the content they share carefully. It enables users to ensure that the shared content genuinely mirrors their personal identity and values. For example, users have the flexibility to disclose intimate and personal information to particular audiences, creating a favorable impression and establishing a friendly self-identity on the OSN platform [33]. This ongoing process of editing and refining content helps users maintain a consistent and authentic self-identity, so that their sense of psychological ownership over the platform is strengthened [58]. Third, it directly supports the need for efficacy and effectance by allowing users to adjust and improve their online interactions after posting content, enhancing their perceived control and competence within the OSN [62]. Users' ability to control and shape content according to their objectives enhances feelings of efficacy, making them perceive the platform as an extension of their capabilities. This perception deepens their sense of psychological ownership [60]. Therefore, we hypothesize the following:

H4. *Editability affordance influences psychological ownership positively.*

3.5. Association affordance and psychological ownership

Association affordance in OSNs captures the technological features that offer users the possibility of associating their posted information with other users who interact with it [38]. This is achieved through actions such as expressing personal opinions using reaction buttons such as "‘‘👍/👎’’", mentioning other users using the "@" button, and tagging posts with the hashtags "#" button [32]. First, this affordance helps users to construct a shared space within OSN that feels familiar and personally significant, where they can express their emotions to others using emoticons, such as "angry," "ha-ha," "wow," "sad," and "love." These association features facilitate meaningful information exchange and develop harmonious relationship bonds among OSN participants [39,54]. By enabling users to associate with others, this affordance helps forge a personalized and secure environment, which enhances users' attachment to the platform and deepens their psychological ownership [55]. Second, it addresses the need for self-identity significantly. It allows users to express who they are and what they stand for by aligning themselves with certain people, topics, or causes, by tagging others and utilizing hashtags to associate with larger communities [32]. Moreover, the affordances aid in the continuous reinforcement and validation of users' self-identities through interactions and responses from the community, thus fostering a deeper sense of psychological ownership. Third, it directly fulfills the need for efficacy and effectance by enhancing users' ability to achieve desired outcomes within the OSN. Through directed associations and hashtags, users can align their content with influential individuals or communities, increasing the visibility and impact of their posts. This capability allows users to leverage their relational ties effectively to enhance their presence and influence within the OSN, thereby enhancing their sense of competence and control over the platform. Consequently, this perceived efficacy reinforces users' psychological ownership, viewing the OSN as a tool they can skillfully use to manage and enhance their social interactions and outcomes [60]. Thus, we hypothesize that

H5. *Association affordance influences psychological ownership positively.*

4. Study 1: online survey

A multimethod approach, comprising a field survey (Study 1) and a scenario-based experiment (Study 2), is employed to examine the research model comprehensively. Study 1 aims to analyze the relationships between various OSN affordances and user self-disclosure behavior in a natural setting. By examining user perception and utilization of these affordances in real-world scenarios, the survey data enhance the research's external validity.

4.1. Sample and data collection

The survey sample targeted current users of WeChat, the most popular OSN platform in China, with nearly 1.15 billion active users [63]. Over 75 % of these users engage with *WeChat moments* daily to share personal experiences or keep up with friends' stories (AdChina, 2022), indicating the significant use of this feature on WeChat [18]. Based on the above, we followed recent OSN self-disclosure studies [13,18,23] to select WeChat as our research context. The survey questionnaire was created and distributed on a third-party professional survey platform, Sojump. Given that WeChat users often follow official accounts, we capitalized on this behavior to reach our target audience by collaborating with specific WeChat official accounts through Sojump.[1] The Sojump official account would forward the survey questionnaire to its subscribers, such as by inserting the URL or QR code of the questionnaire in its posts. Subscribed WeChat users received the survey invitation and had the choice to participate voluntarily. As this study investigated the impact of OSN affordances on self-disclosure, we provided contextual descriptions of accessibility, information retrieval, editability, and association affordances to help respondents have a clear understanding of OSN affordances while answering the survey questions. To encourage respondents to complete the survey, we offered a $3 economic incentive for submitting complete information. A total of 456 valid survey questionnaires were collected, as shown in Table C1 (see Appendix C), with 48.2 % male and 51.8 % female respondents. Furthermore, 99.3 % of respondents were twenty years or older, 80.0 % held a bachelor's degree or higher, and 65.1 % reported a yearly income of $5000 or more. In terms of usage experience and frequency, 89.6 % of respondents had used WeChat for three years or more, and 64.7 % reported using WeChat at least twice per month.

We determined the minimum sample size using G*Power software prior to conducting formal analysis. Both prior and post hoc power analyses were performed for 12 predictors, which included seven control variables. In G*Power, an effect size f^2 of 0.25 and a significance level (α) of 0.05 were selected, with a power of 0.95 (1 - β) probability. Prior analysis indicated that a minimum sample size of $N = 115$ would be required to achieve a power of 0.95. Subsequently, a post hoc analysis utilizing our actual sample size ($N = 456$) demonstrated an actual power around 1.0, surpassing the priori estimation.

We estimated common method bias (CMB) using three statistical

[1] Sojump offers two additional options for researchers to collect data. First, researchers can distribute the questionnaire's URL or QR code to target respondents, an approach commonly employed when researchers are focusing on a specific research group, such as employees within a company. Second, researchers can use the Sojump data mart for data collection purposes. As a professional survey platform, Sojump boasts a large and diverse pool of registered respondents. Furthermore, Sojump equips researchers with various data-filtering tools to recruit eligible respondents. Typically, researchers are required to provide an incentive for participant involvement and incur fees for utilizing Sojump's data collection services.

techniques: Harman's single-factor test [64], the correlation marker technique [64], and the partial least squares (PLS) marker variable approach [65]. First, Harman's single-factor test resulted in the generation of eight factors, explaining 74.39 % of the total variance, with the primary factor contributing only 16.87 % of the variance. This initial analysis indicated a low likelihood of CMB in the survey data.

Second, the correlation marker technique was applied, employing a four-item scale for work exhaustion adapted from Ahuja et al. [66] as a marker variable. The findings revealed that all significant construct correlations maintained significance, and none of the nonsignificant construct correlations became significant, suggesting a minimal presence of CMB.

Third, we used the LS marker variable approach. Step 1 involved selecting four items related to work exhaustion as the marker indicators. In Step 2, we computed method variance correlations and calculated the average correlation between the main items and the marker indicators. In Step 3, we executed the PLS baseline model comprising all primary constructs. In Step 4, we added the marker variable into the baseline model and ran the marker variable model. In Step 5, we compared the PLS baseline model and the marker variable model in terms of path coefficients and statistical significance. In Step 6, we adjusted the cross-loading matrix using the mean correlation between the main items and the marker indicators. The results showed that the size and significance of path coefficients and cross-loading matrix were not changed significantly, alleviating the concern with CMB (see Appendix D).

4.2. Measures

Table C2 presents the constructs and scales. All of the scales were adapted from previously validated measures in the existing literature. We operationalized all scales of principal constructs using a seven-point Likert scale, ranging from "1 = strongly disagree" to "7 = strongly agree." To ensure content validity, we asked subject matter experts to help us review the modified scales before being accepted as the final scales [67].

OSN affordances. We adapted the twelve-item scale from Chan et al. [38] to measure the four types of OSN affordance, namely, accessibility affordance, information retrieval affordance, editability affordance, and association affordance. These OSN affordances were conceptualized as first-order reflective constructs. Chan et al. [38] originally developed the scales for OSN affordances to reflect users' perceived action possibilities offered by multiple technological features of the OSN platform in the IS discipline. Accordingly, we referred to Chan et al.'s [38] scales to measure the four types of OSN affordances.

Psychological ownership. We adapted the fourteen-item scale from Karahanna et al. [31] to measure psychological ownership. Psychological ownership was conceptualized as a third-order formative construct with three dimensions: need for having a place, need for self-identity, and need for efficacy and effectance. Meanwhile, need for having a place and need for efficacy and effectance were framed as first-order reflective constructs. Need for self-identity was modeled as a second-order formative construct with three elements, namely, coming to know the self, expressing self-identity, and maintaining continuity of self-identity. Karahanna et al. [31] originally developed the scales for psychological ownership needs to describe users' feelings of ownership toward social networking sites in the IS discipline. In this study, we extend Karahanna et al.'s [31] scales to capture users' psychological ownership toward OSNs.

Self-disclosure. We adapted the twenty-one-item scale from Zhang et al. [3] to measure self-disclosure. Self-disclosure was operationalized as a second-order formative construct with five dimensions: amount, depth, honesty, intent, and valence. The scales for self-disclosure were originally proposed by Posey et al. [35] in the IS research community. Zhang et al. [3] further extended Posey et al.'s [35] scales to capture users' self-disclosure on the social networking site. In this study, we used the scales of Zhang et al. [3] scales to measure users' self-disclosure on the OSN platform.

Control variables. We included a series of demographic variables, such as age, gender, education, income, usage experience, and usage frequency, as control variables in the model. We also used privacy concerns as a control variable because they were found to influence users' OSN self-disclosure significantly [1,13]. Gender was measured using a two-point scale, with "1 = male" and "2 = female." Age was measured using a four-point scales, from "1 = twenty years or below" to "4 = forty-one years or above." Education was measured on a three-point scale: "1 = below college," "2 = bachelor's degree," and "3 = master's degree or above." Income was measured by users reporting their yearly income in thousands of dollars. Usage experience was measured in months using a six-point scale, from "1 = one year or below" to "6 = five years or above." The five-item scale for privacy concerns was adapted from Ozdemir et al. [68].

4.3. Results

The research model was validated using the PLS structural equation modelling (PLS-SEM) approach. The specific software package was Smart-PLS 3.27. PLS-SEM has advantages in developing theories in exploratory research and focuses on maximizing the explained variance of endogenous variables [69]. Moreover, PLS-SEM works effectively in modeling complex models with both reflective and formative constructs, and does not produce inadmissible solutions and factor indeterminacy problems that can occur with covariance-based SEM approaches [70].

As psychological ownership and self-disclosure were high-order formative constructs, we followed Polites and Karahanna's [71] three-steps procedure to generate latent variables for each of their first-order dimensions. In Step 1, we ran the full model in PLS with all principal first-order reflective constructs. We generated three latent variables for need for self-identity and five latent variables for self-disclosure, corresponding to the numbers of their dimensions. The resulting latent variables for each dimension were then used as measures of the aggregated need for self-identity and self-disclosure. In Step 2, we conducted the full model in PLS with six first-order reflective constructs (accessibility affordance, information retrieval affordance, editability affordance, need for having a place, and need for efficacy and effectance) and two second-order formative constructs (need for self-identity and self-disclosure) to create latent variables for psychological ownership. This yielded three latent variables for psychological ownership, which were then used as measures of aggregated psychological ownership. In Step 3, we estimated the full model in PLS with four first-order reflective constructs (accessibility affordance, information retrieval affordance, editability affordance), one second-order formative construct (self-disclosure), and one third-order formative construct (psychological ownership).

Measurement model. To assess the psychometric properties of reflective constructs, we conducted confirmatory factor analysis to examine convergent validity and discriminant validity [72]. As shown in Tables C3 and C4, the convergent validity of the scales was confirmed, as (1) all item loadings were higher than 0.70, (2) all average variance extracted values (AVEs) were greater than 0.50, and (3) all composite reliability (CR) values were at 0.70 or above [73]. Moreover, the discriminant validity of the scales was supported because all square roots of AVEs were greater than construct correlations [73]. To test the psychometric properties of formative constructs, we generated formative indicators to examine weights, loadings, and multicollinearity. As shown in Table C5, all weights and loadings for formative indicators were statistically significant. Meanwhile, all variance inflation factor values (VIFs) for formative indicators were lower than 3.30, thereby suggesting minimal possibility of multicollinearity [74,75]. Before proceeding with the structural model analyses, we assessed model fit using two key model fitting parameters [76,77]: the standardized root mean square residual (SRMR) and the root mean square residual covariance (RMS_{theta}). These two statistical parameters indicated that

the data fit the model well ($RMS_{theta} = 0.130$, SRMR = 0.043).

Structural Model. The structural model was tested using PLS-SEM. As shown in Fig. 3, psychological ownership (β = 0.61, $p < 0.001$) was associated positively with self-disclosure, thereby supporting H1. Meanwhile, accessibility affordance (β = 0.18, $p < 0.001$), information retrieval affordance (β = 0.32, $p < 0.001$), and association affordance (β = 0.25, $p < 0.001$) were related positively to psychological ownership, thereby confirming H2, H3, and H5. Contrary to our expectations, editability affordance had nonsignificant effects on psychological ownership (β = –0.03, $p > 0.05$), thereby rejecting H4. In terms of control variables, gender (β = –0.12, $p < 0.01$), education (β = –0.11, $p < 0.01$), and usage frequency (β = 0.17, $p < 0.001$) exerted significant influence on psychological ownership, whereas usage frequency (β = 0.29, $p < 0.001$) had a positive effect on self-disclosure. The overall model explained 44 % and 57 % of the variance in psychological ownership and self-disclosure, respectively.

5. Study 2: scenario-based experiment

While Study 1 captured the behavior and perceptions of real-world users, enhancing the external validity of the research, it merely establishes correlational relationships between different OSN affordances and user self-disclosure. Moreover, self-reported data from users may introduce biases such as social desirability or recall bias. Therefore, Study 2, a scenario-based experiment, was conducted to offer additional insights that go beyond the scope of Study 1. The experimental design facilitated the testing of specific hypotheses under controlled conditions, allowing the isolation of crucial variables and the examination of causal effects that might be difficult to disentangle in a survey setting. Additionally, the controlled experimental setting helps reduce potential biases inherent in survey data by observing actual behavior rather than relying solely on self-reports.

5.1. Sample and data collection

The scenario-based experiment uses vignettes to present participants with written descriptions of realistic situations, followed by requests for responses on various rating scales that measure the variables of interest. This experimental approach is commonly employed in self-disclosure literature [16,78,79]. By offering subjects contextual and situational details using hypothetical terms, the scenario-based experiment enhances the realism of the decision-making process [80]. In Study 2, we used a scenario-based OSN platform, specifically *WeChat*, incorporating virtual manipulations through a 2 × 2 × 2 × 2 factorial design to apply treatments and validate the research model. Consistent with Study 1, the primary focus was in the context of *WeChat Moments*. The scenario-based experiment was conducted via an online survey platform operated by Credamo, a professional Chinese survey platform extensively used in China-based studies [81,82]. This platform facilitated the implementation of manipulations through pictures of *WeChat Moments* accompanied by textual scenario descriptions and the random distribution of survey participants.

Our experimental procedures encompass three distinctive stages. First, all participants were briefed on the academic nature of the experiment and informed that their involvement was voluntary, with assurance that only aggregate data excluding personal identifying information would be used. To incentivize attendance, respondents were offered ¥5 (about $0.70) for valid and complete responses. Second, participants received a uniform scenario in which they envisioned themselves as users of the social networking service *WeChat Moments*, following which, one of sixteen experimental scenarios was randomly allocated to each respondent (see Fig. 4). Third, a manipulation check was conducted to gauge the extent of OSN affordances in relation to the designated scenario. Meanwhile, participants were required to complete the follow-up questionnaire based on the assigned scenario, which includes their perceptions regarding psychological ownership, self-disclosure, and privacy concerns, as well as their demographics (e.g., age, gender, income, education, and usage frequency). For a full description of the experiment, please see Appendix E.

As shown in Fig. 4, for each type of affordances, we developed "low" and "high" conditions in alignment with a 2 × 2 × 2 × 2 experimental design, employing a randomized and intersecting between-subjects approach. The four types of OSN affordances are as follows: (1) accessibility affordance (low versus high), (2) information retrieval

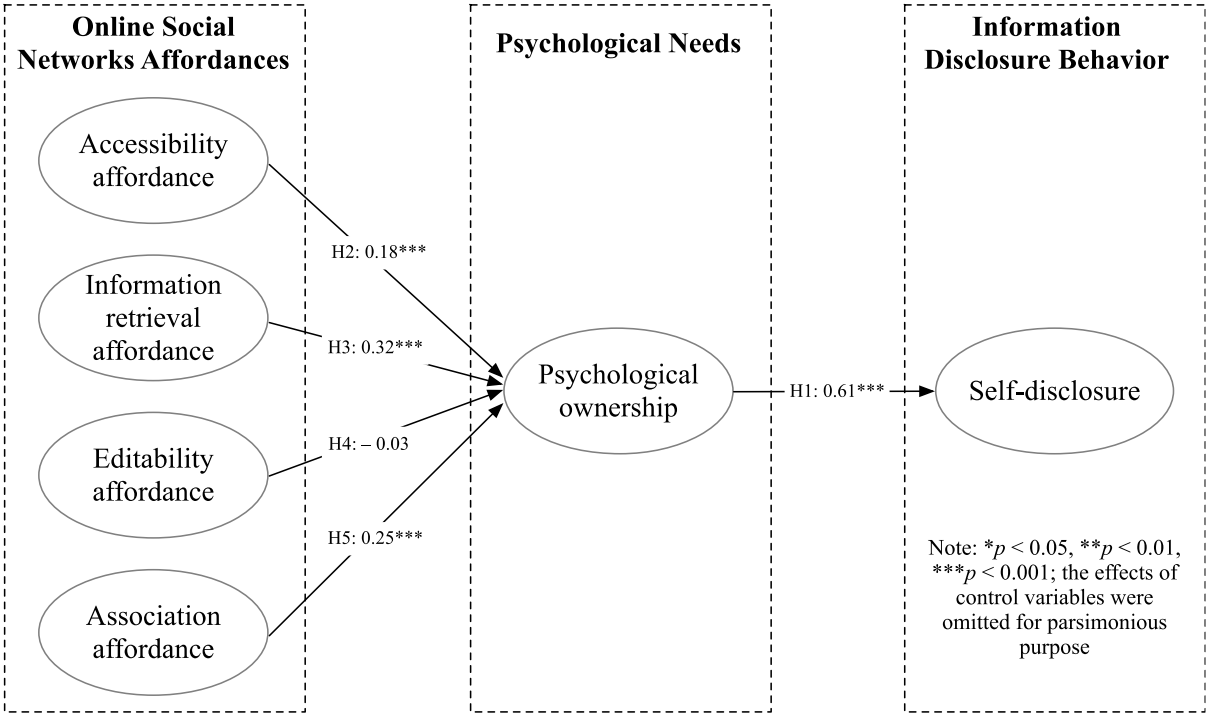


Fig. 3. Structural model.

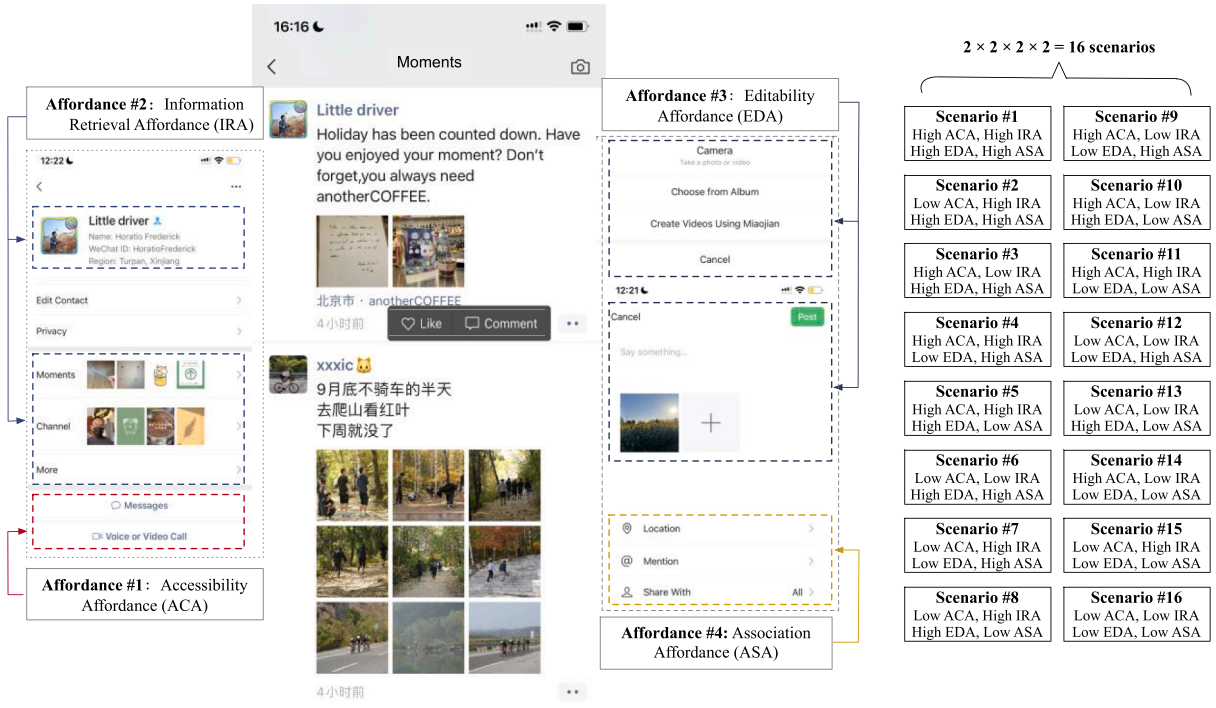


Fig. 4. Screenshots of WeChat Moments, affordances, and scenarios.

affordance (low versus high), (3) editability affordance (low versus high), and (4) association affordance (low versus high), culminating in a total of 16 experimental scenarios. To enhance the contextual validity and realism, these experimental scenarios were refined based on feedback received from expert users with rich OSN usage experience. Furthermore, a pretest involving 20 *WeChat Moments* users was conducted to validate the manipulation of these experimental scenarios for relevance and comprehensibility. We followed established procedures to ensure the data quality for online data [80]. From the 654 WeChat users who agreed to participate in the experiment, we gained 580 valid responses (effective response rate of 88.69 %); 74 responses were excluded for various reasons such as respondent haste during questionnaire completion or incomplete answers. To establish the minimum required sample size for our study, power analyses were conducted using G*Power [83]. The analysis revealed that a total of 580 participants would ensure a statistical power of 0.99 for a medium effect size (f^2 = 0.25).

5.2. Measures

To ensure construct validity, all measures for principal variables were adapted from prior studies and tailored to the context of self-disclosure on OSNs. All survey items were measured by a 7-point Likert scale, ranging from 1 (strongly disagree) to 7 (strongly agree). We followed the commonly accepted advice on wording questions when developing and finalizing the measures.

OSN affordances. We operationalized each of the four OSN affordances with two levels: low and high. Therefore, accessibility (information retrieval/editability/association) affordance was coded as follows: 0 = if the respondents were provided with the low-accessibility (information retrieval/editability/association) experimental scenario and 1 = if the respondents were provided with the high-accessibility (information retrieval/editability/association) experimental scenario.

Psychological ownership. We measured psychological ownership using the 14-item scale from Karahanna et al. [31] as in Study 1. We employed a seven-point Likert scale for all items of psychological ownership. Therefore, psychological ownership was coded as follows: 1 = if the respondents strongly disagree that they have a sense of psychological ownership toward *WeChat Moments* and 7 = if the respondents strongly agree that they have a sense of psychological ownership toward *WeChat Moments*.

Self-disclosure. We measured self-disclosure using the 21-item scale from Zhang et al. [3] as in Study 1. We operationalized all items of self-disclosure using a seven-point Likert scale. Therefore, self-disclosure was coded as follows: 1 = if the respondents strongly disagree that they have a willingness to disclose personal information toward *WeChat Moments* and 7 = if the respondents strongly agree that they have a willingness to disclose personal information toward *WeChat Moments*.

Control variables. We collected participants' demographic information as control variables. These demographic variables included age, gender, education, income, and usage frequency as in Study 1. Meanwhile, we adapted a three-item scale from Ozdemir et al. [68] to measure privacy concerns as in Study 1, for the purpose of conducting robustness checks. We measured privacy concerns by using a seven-point Likert scale. Specifically, privacy concern was coded as follows: 1 = if the respondents strongly disagree that they are concerned about the potential loss of privacy as a result of information disclosure toward *WeChat Moments,* and 7 = if the respondents strongly agree that they are concerned about the potential loss of privacy as a result of information disclosure toward *WeChat Moments*.

5.3. Results

Subject background information. The demographic information of subjects is shown in Table F1 (see Appendix F). Among the 580 participants in the experiment, there were 258 males (44.5 %) and 322 females (55.5 %). The majority of participants (86.7 %) fell within the age range from twenty to forty. Furthermore, 442 participants (76.2 %) held a bachelor's degree or higher, and 367 participants (63.3 %) reported a monthly income exceeding ¥5000 (approximately $690). An evaluation of nonresponse bias was conducted to compare the usage experience and age of early and late respondents, with results showing no significant differences ($p > 0.05$), indicating minimal non-response bias.

Manipulation check. Prior to hypothesis testing, we conducted an analysis of variance (ANOVA) test to assess the manipulations regarding

accessibility (information retrieval/editability/association) affordance. Specifically, we asked participants to retrieve their usage experiences to assess the level of accessibility (information retrieval/editability/association) affordance of *WeChat moments* using adapted OSN affordance scales. As shown in Table F2, the mean rating in the low-accessibility affordance (i.e., $mean_{low}$) was 1.25, and the mean rating in the high-accessibility affordance (i.e., $mean_{high}$) was 6.64, indicating a successful experimental manipulation ($F = 10{,}490.59$, $p < 0.001$). Similarly, manipulation of information retrieval affordance ($mean_{low} = 1.38$, $mean_{high} = 6.53$, $F = 6342.06$, $p < 0.001$), editability affordance ($mean_{low} = 1.34$, $mean_{high} = 6.58$, $F = 6762.37$, $p < 0.001$), and association affordance ($mean_{low} = 1.32$, $mean_{high} = 6.56$, $F = 7019.12$, $p < 0.001$) was also successful.

Hypotheses testing. First, we used the OLS regression model with SPSS 22.0 software to test the hypothesis regarding the effects of psychological ownership on self-disclosure. The findings revealed a positive influence of psychological ownership on self-disclosure ($\beta = 0.876$, $p < 0.001$), corroborating H1. Second, an ANOVA test was conducted using SPSS 22.0 to test the effects of OSN affordances on psychological ownership. As shown in Fig. 5a, psychological ownership was significantly higher in the high-accessibility affordance condition than in the low-accessibility condition ($M_{highACA} = 4.51$ vs. $M_{lowACA} = 3.76$, SDs = 1.50 and 1.51, $F = 36.10$, $p < 0.001$), providing additional evidence for H2. Likewise, as shown in Figs. 5b, 5c, 5d, the ANOVA results show that information retrieval affordance ($M_{highIRA} = 4.27$ vs. $M_{lowIRA} = 3.99$, SDs = 1.55 and 1.54, $F = 4.76$, $p < 0.05$), editability affordance ($M_{highEDA} = 4.77$ vs. $M_{lowEDA} = 3.51$, SDs = 1.32 and 1.51, $F = 114.54$, $p < 0.001$), and association affordance ($M_{highASA} = 4.91$ vs. $M_{lowASA} = 3.54$, SDs = 1.35 and 1.42, $F = 139.97$, $p < 0.001$) were positively associated with psychological ownership, thereby confirming H3, H4, and H5.

6. Discussion and conclusions

The primary objective of this research was to examine the formation of user self-disclosure in the context of OSNs. Drawing on affordance theory [34] and the NAF perspective [31], this research investigated how OSN affordances influence user self-disclosure through the formation of psychological ownership. Both a field survey and a scenario-based experiment were conducted to validate the research model empirically, with meta-inferences drawn from a synthesis of the survey and experiment studies. As shown in Table F3, psychological ownership has a positive effect on self-disclosure in both survey and experimental settings, thereby supporting H1. Moreover, accessibility, information retrieval, and association affordances were positively

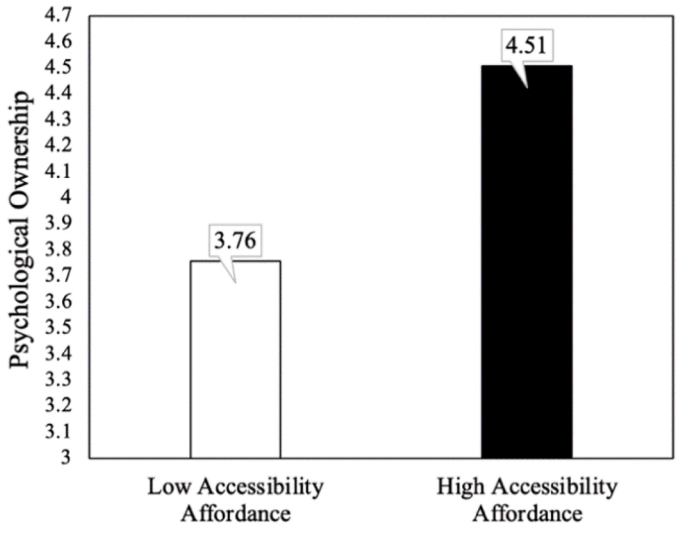


Fig. 5a. The effect of accessibility affordance.

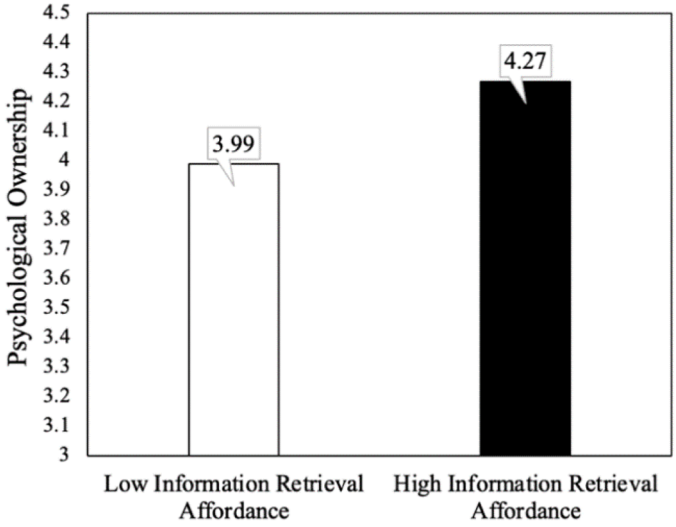


Fig. 5b. The effect of information retrieval affordance.

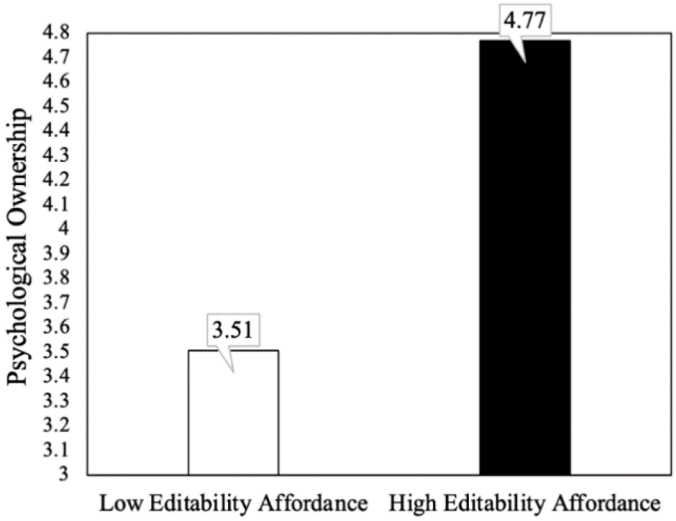


Fig. 5c. The effect of editability affordance.

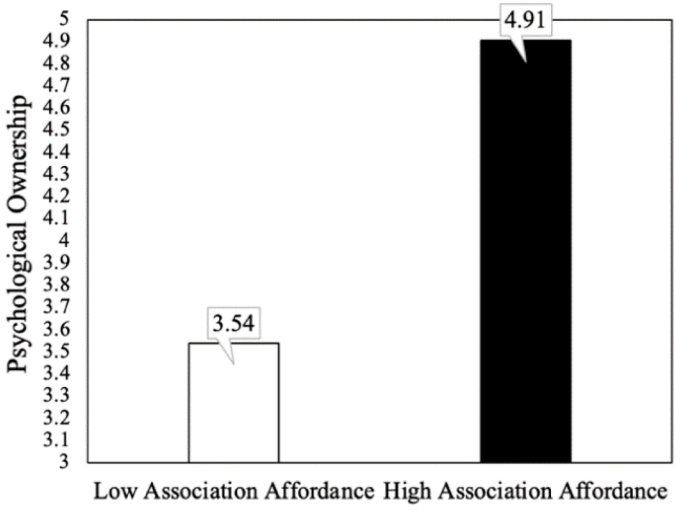


Fig. 5d. The effect of association affordance.

related to psychological ownership in both survey and experimental studies, thereby supporting H2, H3, and H5. In contrast, while the editability affordance exhibited no significant influence on psychological ownership in the survey, it had a significant effect in the experiment, partially supporting H4. Additionally, in the survey sample, psychological ownership mediated the effects of accessibility, information retrieval, and association affordances on self-disclosure, whereas it had no mediating role in the relationship between editability affordance and self-disclosure. Conversely, in the experimental sample, psychological ownership acted as a mediator for the effects of accessibility, information retrieval, editability, and association affordances on self-disclosure.

Based on the empirical results discussed, it is apparent that the impact of the editability affordance on psychological ownership differed between Study 1 and Study 2. Two plausible explanations could elucidate this distinction, viewed through the lens of allocentric and egocentric affordances alongside methodological differences inherent in a survey-based study versus a scenario-based study. First, allocentric affordances, such as accessibility, information retrieval, and association, are inherently social and involve interactions with others [31]. These affordances actively engage users in social connections and content exchange, which are central to their experiences on OSNs. Consequently, users may naturally attribute a sense of psychological ownership to OSNs when these affordances are in play, facilitating a more profound integration of the user's social identity within the network. In contrast, editability affordance, an egocentric affordance, emphasizes individuals' ability to control and personalize their own content [31]. It prioritizes autonomy and personal control rather than social interaction. In a survey context, where participants reflect on their habitual use of OSNs, the editability affordance may not be as prominent or influential in fostering a sense of psychological ownership. Users may place more importance on the social aspects of the platform, rendering the personal control afforded by editability secondary or less influential in their overall experience.

Second, the survey method captures participants' general perceptions and behavior in their natural, everyday use of online social networking platforms. In this context, users tend to perceive the platform primarily through the lens of its allocentric affordances, which dominate their typical experience. Thus, the lack of a significant effect of the editability affordance on psychological ownership in the survey may reflect its peripheral role in users' day-to-day engagement with the platform. In contrast, in the scenario-based experiment, participants are presented with specific scenarios emphasizing distinct affordances. When participants are prompted to engage with and manipulate their own content, the editability affordance becomes more prominent, heightening users' awareness of their content control and ownership. This heightened focus can lead to a stronger perception of psychological ownership, as the experimental design highlights an affordance that might otherwise be overlooked in routine use.

6.1. Theoretical contributions

This paper contributes to the literature significantly in three main ways. First, it enriches the self-disclosure literature by shedding light on the influence of contextual technological factors on users' OSN self-disclosure. While previous studies focused primarily on examining the antecedents of OSN self-disclosure pertaining to generic sociopsychological factors, such as self-presentation, relationship maintenance, privacy risks, privacy concerns, and emotional reactions (see Appendix A), the impact of contextual technological factors has largely been neglected. By shifting the current focus of antecedents of OSN self-disclosure from generic sociopsychological factors to contextual technological factors such as accessibility, information retrieval, editability, and association affordances, our study enriches the *behavioral* perspective on OSN self-disclosure, with these relationships being both theoretically justified and empirically verified.

Furthermore, our research contributes to the field by exploring the effects of psychological ownership on users' self-disclosure. In the IS research community, psychological ownership has been studied as a determinant of various technology use behavior, such as technology adoption and use [84], information security policy compliance [85], and crowdfunding participation [86]. However, limited attention has been given to the impact of psychological ownership on users' OSN self-disclosure. The OSN platform offers users a virtual "home" territory to establish relationship connections with other users [10]. Such virtual "home" territory satisfies users' psychological needs for having a place, expressing self-identity, and maintaining efficacy and effectance [31].

Second, our work advances affordance theory by conceptualizing contextual OSN affordances, namely, accessibility, information retrieval, editability, and association. While previous studies in IS have begun applying affordance theory to understand technology use behavior in OSNs, only a few studies provided a theoretical topology and operational measures of OSN affordances (see Appendix B). In particular, Chen et al. [39] developed the topology and measures of OSN affordances to examine employees' enterprise social media use, whereas Chan et al. [38] proposed the topology and measures of OSN affordance to investigate users' social networking site cyberbullying. Aligning with Chan et al.'s [38] typology of OSN affordances, we elaborate on four archetypes of OSN affordances, progressively enhancing the operationalization and testing potential of affordance theory.

Moreover, our work extends the NAF perspective by validating empirically that psychological ownership plays a crucial mediating role in the relationship between OSN affordances and users' self-disclosure. According to the NAF perspective, individuals are motivated to utilize OSN applications based on their intrinsic psychological needs, particularly when these applications offer various affordances that cater to these needs [31]. Essentially, the technological features of OSNs present diverse technology affordances that enable users to satisfy their intrinsic psychological needs and enable goal-oriented users to engage in diverse technology use behaviors [38,40]. However, the underlying mediating mechanisms between technology affordances and technology use behaviors is lack of investigation [31]. Our work pioneers the exploration of how OSN affordances influence users' self-disclosure through the lens of psychological ownership, significantly advancing the NAF perspective. By examining how OSN affordances impact the subdimensions of psychological ownership and user self-disclosure behavior, our research offers a comprehensive understanding of how users' various psychological needs mediate the effects of technology affordances on their technology use behaviors.

6.2. Practical implications

Prior studies on OSN self-disclosure has demonstrated the influence of sociopsychological factors on users' inclination to disclose personal information on the OSN platform [10,13,23]. While these insights are valuable, our empirical findings provide actionable guidance to OSN practitioners on how to intentionally design multiple affordances and implement proactive measures to enhance users' self-disclosure, with a focus on OSN psychological ownership and technological features.

First, OSN providers should strategically foster users' sense of psychological ownership to encourage voluntary information disclosure. By addressing the conceptualization and dimensions of psychological ownership, OSN providers are recommended to satisfy users' psychological needs for having a place, expressing self-identity, and maintaining efficacy and effectance. For example, the OSN platform could incorporate anthropomorphic emoticons and avatars to enhance users' feelings of territory. Moreover, the OSN platform could introduce activities such as "footprints of the past," enabling users to compile mementoes, photographs, videos, diaries, and gifts to document their past experiences, which can be shared with close friends and family. These activities aid users in defining themselves and articulating their identities. Additionally, the OSN platform could offer "social cooperation and competition" features, allowing users to engage in challenges that match

their abilities and seek opportunities for skill enhancement, knowledge acquisition, competency development, or a sense of accomplishment.

Second, OSN providers should direct their attention to strategically improving users' awareness and use of OSN affordances. Based on the topology and measures of OSN affordances, OSN providers are advised to design actionable affordances, such as accessibility, information retrieval, editability, and association, intentionally. For example, OSN providers can design technological features to enhance accessibility functions, such as friending, following, and chatting. Moreover, OSN providers can design technological features to facilitate information retrieval functions, such as searching and browsing. Meanwhile, OSN providers can design technological features to promote editability functions, such as erasing, editing, and hiding. Additionally, OSN providers can design technological features to increase association functions, such as directed associations ("@") and hashtags ("#").

6.3. Limitations and future research

This research has several limitations and paves the way for future research opportunities. First, in line with the NAF perspective, our work focuses solely on exploring the mediating role of psychological ownership in the relationships between OSN affordances and user self-disclosure. Some of our research findings align with those of Zhao et al. [87], suggesting that in scenarios of complementary and competitive mediations, the significant direct effect (*c*) might indicate the existence of an unexplored secondary mediator warranting investigation in future studies. For instance, future research could leverage social exchange theory (e.g., [16,88]) to examine the mediating roles of perceived privacy invasion and perceived relationship bonding, which can transmit the effects of OSN affordances on user self-disclosure, given the significance of both privacy invasion and relationship building in OSNs. Additionally, subsequent research endeavors could employ affect heuristic theory (e.g., [37]) to investigate how passive or active emotions mediate the relationships between OSN affordances and user self-disclosure.

Second, another future direction is cross-context theory contextualization. According to Hong et al. [28], cross-context theory contextualization allows researchers to replicate a theoretical model in different contexts and consolidate findings into a context-contingent theory. Future research should enhance the generalizability of the research findings by extending our theoretical model from China to other cultural regions, such as Canada, the United Kingdom, and the United States. This approach helps to validate the assumption of generalizability and identify contingency factors of relationships of our theoretical model across contexts. For example, collectivistic cultural regions often maintain a concern for the well-being of the group and a tendency to group cooperation [89]. In contrast, individualistic cultural regions are influenced largely by personal preferences and cost–benefit analyses [89]. Therefore, allocentric affordances (e.g., accessibility, information retrieval, and association) may more significantly affect psychological ownership and self-disclosure in collectivistic cultures, while egocentric affordances (e.g., editability) may more significantly influence psychological ownership and self-disclosure in individualistic cultures. Besides, future research should facilitate the generalizability of the research findings by extending our theoretical model from private social networks (e.g., WeChat) to public social networks (e.g., Weibo). These two types of social networks exhibit significant differences that could influence the formation of users' self-disclosure behavior.

Third, while our research incorporates both a field survey and a scenario-based experiment to provide robust findings with enhanced external and internal validity, notable inconsistencies in the impact of the editability affordance are evident between Study 1 and Study 2. This discrepancy presents a potential limitation but also signifies a promising avenue for future research. The differing effects observed in the survey and scenario-based experiment highlight the critical aspect of the nature of the affordance—whether allocentric or egocentric—and its alignment with the specific research context. Future research should explore how varying methodological approaches may emphasize these affordances differently, potentially skewing their perceived influence on psychological ownership. Investigating this dynamic across diverse contexts and user experiences could yield profound insights into the roles of affordances in shaping user perceptions and behavior.

6.4. Conclusions

The collection of personal information from users is an unavoidable element of the development of OSN platforms. However, how to promote users' self-disclosure remains the most challenging issue in the OSN platform. Drawing on affordance theory [34] and the NAF perspective [31], we develop a model to explain the formation of users' self-disclosure in the OSN platform. Our empirical results show that accessibility, information retrieval, editability, and association affordances positively influence psychological ownership, which further formulate users' OSN self-disclosure. Additionally, psychological ownership mediates the effects of accessibility, information retrieval, editability, and association affordances on users' OSN self-disclosure. We believe that these empirical findings have significant implication for research on self-disclosure and provide practical guidance to facilitate self-disclosure in the OSN platform.

CRediT authorship contribution statement

Zhenxin Xiao: Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Xiang Gong:** Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Christy M.K. Cheung:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

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Supplementary materials

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